

**Introduction to Engineering Design with Professional Development I (ENGR
2050) Classroom Lab Activity**

19-4	Professional and Societal Considerations Assignment
Textbook Reference:	N/A
Purpose / Goal:	To clarify student thinking beyond the technical aspects required to develop their project to include
Materials / Resources Required:	Project design process materials
Time Allocated:	Take Home Assignment

Introduction & Instructions

As part of your design process, your team must make informed judgments that consider the impact of engineering solutions in global, economic, environmental, and societal contexts. In the following sections, describe how you **individually / personally** considered this and provide your evaluation.

Each student is to *individually* complete this form for their **overall team project**, *not* their individual subsystems. Note that the impacts in each area may be positive or negative.

After each section, indicate your evaluation of the impact for each topic. Indicate a strong **positive** impact as a 5, no impact as a 3 and a strong **negative** impact as a 1.

ALL SECTIONS ARE REQUIRED!

Student Name	Peijing Xu			
Team Name	Iron Volt Mobility			
Section Number	5	RCSID	xup2	@rpi.edu

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Public Health and Safety

Impact Description	The product's potential positive impact on health and safety includes improving the customer's mobility and mental health, along with promoting a sense of belonging and community integration and the physical health benefits that come with more interactions. Having more independence when transitioning from a walker to a wheelchair and vice versa can increase the physical exercise the customer is getting and make the customer more active daily. Some studies show that physical mobility improves both the physical and mental health of everybody, not just those who are tragically inflicted with a disability. For people with disabilities, studies show that having more independence in their daily lives can reduce depression and anxiety thus improving their mental health [1]. Being more active would mean that the customer would likely be able to participate in more social activities with his or her peers to create a sense of belonging as well as a community where they could thrive.
Impact Evaluation Strong <i>positive</i> impact=5 No impact = 3 Strong <i>negative</i> impact =1	5

Impact Description	The product has a positive impact globally because it can influence people around the world. The product can foster a more connected and inclusive global community where those with disabilities can have more independence and inclusivity. Our product creates the opportunity for mutual learning between those with and those without disabilities fostering growing compassion for the disabled and modesty for those without[2]. The product has the potential to improve the quality of life for many far beyond just the United States and could reach a global scale if successful enough. The increased community inclusion and empathy provided by the product could lead to a widespread trend of inclusion for the disabled where communities that had embraced these changes could serve as models for other communities. Eventually, the trend could cause shifts in global views on the disabled and encourage the allocation of resources towards them like with eco-friendly products and sustainability movements.
Impact Evaluation Strong <i>positive</i> impact=5 No impact = 3 Strong <i>negative</i> impact =1	4

Cultural

Impact Description	Our product has a positive cultural impact because giving more mobility and independence to those with disabilities can curb some resentment and discrimination against disabled people in some cultures. Many cultures around the world look down on and discriminate against those with disabilities[3], with the primary reason being that they believe that typical abilities are superior to them[4], but our product allows for more mobility and independence to bring disabled people more into the community, which would foster more compassion and humility while breaking down physical and social barriers[2]. As the disabled begin to become more active in society, participating and contributing to community life, the stereotype that those without disabilities are superior becomes challenged and can be more easily broken down. This could cause cultures that traditionally looked down upon those with disabilities to shift towards more inclusion and acceptance for those with disabilities.
Impact Evaluation Strong <i>positive</i> impact=5 No impact = 3 Strong <i>negative</i> impact =1	4

Societal

Impact Description	The product will only transition a disabled person safely from standing to sitting without much trouble and would not have any meaningful impact on a society aside from helping disabled youth be more independent in their day-to-day lives. However, there still are some benefits of more community inclusion for a disabled person like how being around those with disabilities allows for the development of more compassion and humility[2]. Having more disabled people around would permit others to see the various struggles that they deal with and a better understanding of how fortunate those who are born without disabilities are. With more independence, those with disabilities could offer other perspectives to today's society and their increased inclusion could cause a societal change to facilitate more resources to create space for everybody to thrive in[2]. In conclusion, there would be some small positive impact on communities where there are customers who utilize the product for more independence, becoming more open-minded and more compassionate.
Impact Evaluation Strong <i>positive</i> impact=5 No impact = 3 Strong <i>negative</i> impact =1	4

Impact Description	Our product is meant to be an attachment to a wheelchair making it easily reusable, just un-attach and reattach it to another wheelchair if the previous wheelchair, for whatever reason, cannot be used anymore, our product can easily be transferred and reused. Having this function will have a decent positive impact on the environment because the product being reusable means less materials will be harvested, which means less energy expended and less pollution produced[5]. The main components of the project would be wood and steel, where the wood is from the platform that the customer can push themselves from and acts as support, while the steel is from the hinges and motors that are prevalent to the product. With the product being reusable, less wood would be extracted leading to a more diverse ecosystem and a decrease in carbon dioxide levels in the atmosphere[6]. Steel would also be extracted less and would contribute to the decrease in carbon emissions as steel production typically produces 3.3 million tons annually of global emissions[7].
Impact Evaluation Strong <i>positive</i> impact=5 No impact = 3 Strong <i>negative</i> impact =1	4

Economic
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Impact Description	There will not be any negative impact on the economy while the product can only slightly benefit the economy. The product is only an attachment and will not affect the price of any wheelchairs or walkers, because our product cannot be used as a replacement for those tools in any shape or form, therefore, it cannot be used to affect the prices of wheelchairs or walkers. The benefits that the product can bring are some positive labor effects generated from our new product[8]. New technology can increase demand for labor[9], thus creating new jobs and opportunities like a manufacturing company to create parts for the product, hiring more workers to assemble the product, hiring people to distribute the product to the buyers, and many more ways. Creating more jobs and opportunities despite being small could prove beneficial to the economy especially when the product is new and does not have any competition or any similar existing product.
Impact Evaluation Strong <i>positive</i> impact=5 No impact = 3 Strong <i>negative</i> impact =1	4

Citations

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5. Reduce and Reuse Basics | Irving, TX - Official Website. (n.d.). Ww.cityofirving.org. <https://www.cityofirving.org/579/Reduce-and-Reuse> Accessed on 11/13/2024
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